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Topological and shape optimization methods for inverse shape reconstruction in elliptic PDEs

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This talk focuses on inverse reconstruction problems governed by elliptic partial differential equations, where the objective is to identify unknown subregions associated with discontinuous coefficients from boundary measurements. We discuss numerical methods based on topological sensitivity analysis and shape optimization for recovering the location and geometry of hidden structures. Particular emphasis is placed on the development of efficient reconstruction algorithms, their numerical implementation, and robustness with respect to noisy measurements. Numerical examples are presented to demonstrate the accuracy and effectiveness of the proposed reconstruction methods.

Part of this talk is based on joint work with L. Afraites (Université Sultan Moulay Slimane, Morocco), A. Hadri (Université Ibn Zohr, Morocco), M. Hrizi (Université de Monastir, Tunisia), and H. Kahlaoui (Université de Tunis el Manar, Tunisia).

Keywords: Geometric inverse problems, topological sensitivity analysis, shape optimization, interface reconstruction

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References